

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 57
Course: CSE 223: Database Management System

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

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- | | | |
|----|--|-------|
| | | 2x4 |
| 1. | a) Insert the following keys in a B+ Tree.
50 100 49 101 40 102 70 104 106 80 108
I. For m=3
II. For m=4 | |
| | b) Why B+ Tree is used in Database Management System? | 2 |
| 2. | a) What are the rules for a relation to be in BCNF? | 2 |
| | b) Find the Candidate Key considering $R=(A, B, C, D, E)$ and $F=\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ | 5 |
| | c) Find the Canonical Cover of F. | 3 |
| 3. | a) Difference between Inner Join and Outer Join. | 2 |
| | b) "Candidate key is a subset of a super key". Is this statement True/False? Then explain the reasons with proper examples | 4 |
| | c) Explain Set-Intersection operation using Set-Difference operation with a diagram. Give some examples of SQL Aggregate functions. | 4 |
| 4. | Customer(<u>cus_no</u> , cus_name, address, telephone, e-mail)
Order(<u>order_no</u> , order_date, cus_no, total_price, delivery_date)
Order line(<u>order_no</u> , <u>prod_no</u> , quantity, line_total, color, price, unit_price)
Product(<u>prod_no</u> , prod_name, prod_type, unit_price, weight, quantity)
Supplier(<u>sup_no</u> , sup_name, prod_no, telephone, city)
Shipment(<u>prod_no</u> , <u>sup_no</u> , quantity, date, total) | |
| | a) Write SQL: | 2x2.5 |
| | I. Find product name for products that either weight more than 50kg or are supplied by supplier SS3. | |
| | II. Find supplier numbers and names for suppliers who do not supply product type sofa or chair. | |

2x2.5

b) Write relational algebra:

- Find all product types which are supplied in the Sylhet city.
- Find all customer names whose product's delivery date is '2024-06-23'

1+3

5. a) Define ER Diagram and explain the ACID properties of a Transaction.

- b) Check whether the given schedule S is conflict serializable or not. If, yes, then determine all the possible serialized schedule-

T1	T2	T3	T4
			R(A)
	R(A)		
		R(A)	
W(B)			
	W(A)		
		R(B)	
	W(B)		

6. a) Define Canonical Cover and Extraneous Attribute.

2

b) Consider that, $R = (ID, Name, Age)$ $F = \{ID \rightarrow Name, Name \rightarrow Age\}$

8

- What are the conditions of 1 NF.
- Check whether R is in BCNF or not.
- If not mention the reason and
- Decompose R in BCNF

Metropolitan University
Department of Computer Science & Engineering
Final Examination - Spring 2024
Program: CSE; Batch: 57th
Course: CSE 211; Digital Logic Design

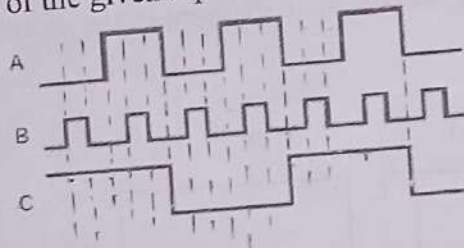
Marks: 40

Time: 2 Hours

[Answer any four of the following questions]

1. (a) Convert the decimal value 4897 in each of the following number systems (you must show calculation). (i) binary (ii) hex (iii) octal (iv) BCD 5

- (b) Draw the output waveform of the given inputs for three input NOR and AND gates. 5



2. (a) Simplify these expressions:

i) $x = \overline{A}C(\overline{A}BD) + \overline{A}B\overline{C}D + A\overline{B}C$

ii) $F_1(A, B, C) = ABC + A'B'C + A'BC + ABC' + A'B'C'$

- (b) Using Karnaugh Map Simplify the following Boolean function F, together with the don't-care conditions d. 5

$$F(A, B, C, D) = \sum(2, 4, 10, 12, 14)$$

$$d(A, B, C, D) = \sum(0, 1, 5, 8)$$

3. (a) Construct a BCD adder using 4-bit binary adder. 8

- (b) Draw the Circuit diagram of a full adder and write the equation for sum and carry output. 2

4. (a) Draw a 3 to 8 line decoder circuit. 4

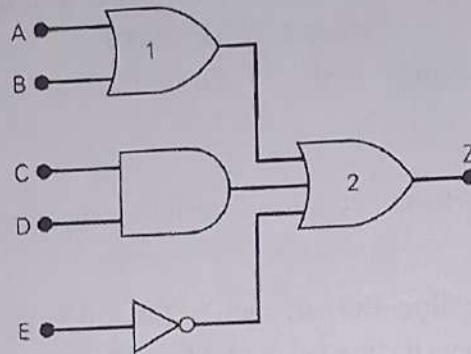
- (b) Implement the following function with a multiplexer: 6

$$F(A, B, C, D) = \sum(1, 3, 4, 11, 12, 13, 14, 15)$$

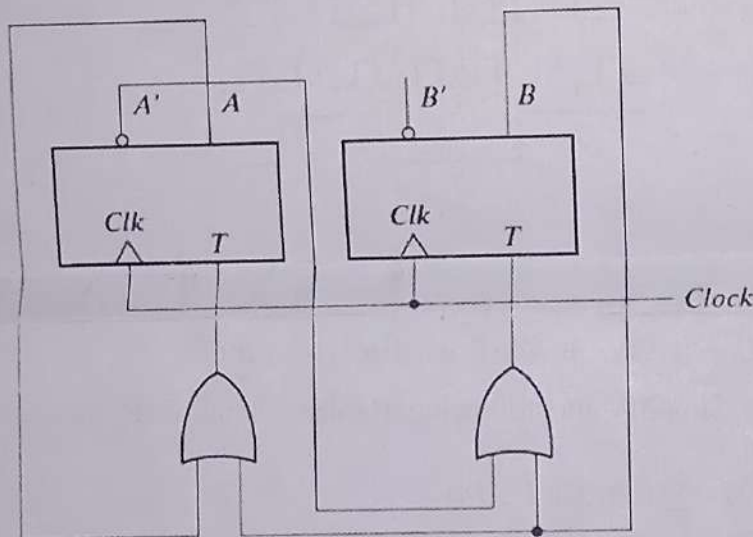
5. (a) Draw the circuit diagram of a JK Flip-Flop. 2

- (b) What is a Register? Design a 4 bit Register with d-type flip-flops. 3

- (c) When the output of the logic circuit in figure goes LOW, it activates another logic circuit, 5
 modify the circuit diagram to represent the circuit operation more effectively. Also
 determine the input conditions (in statement/sentence) necessary to activate the circuit.



6. (a) Show the operation of a basic Latch circuit. 4
 (b) Derive the state table and the state diagram of the sequential circuit shown below. 6



Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 57
Course: CSE 327: Theory of Computation

Marks: 40

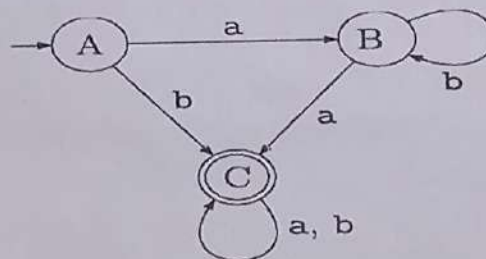
Time: 2 Hours

Answer any **four sets** of questions. All parts of each question must be answered sequentially.
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4 × 1.5 = 6

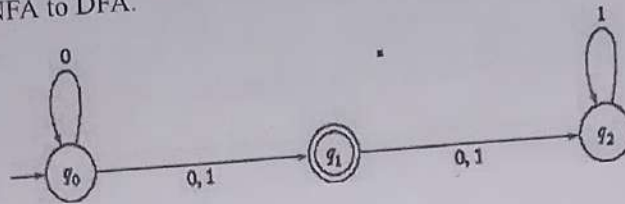
1. a) Find Regular Expressions for the following regular languages:
- $L_1 = \{vwv : w \in \{a, b\}^*, |v| = 2\}$
 - $L_2 = \{\text{Language accepting strings of length at most 2}\}$ where $\Sigma = \{a, b\}$
 - $L_3 = \{\text{All string having even number of 0's}\}$ where $\Sigma = \{0, 1\}$
 - $L_4 = \{\text{Accepting all the strings in which any number of b's follows one or more number of a's is followed by any number of c's}\}$
- b) What are the differences between regular grammar and context-free grammar? 2
- c) Construct right-linear grammars for the following language: 2
 $L = \{a^n b^m : n \geq 2, m \geq 3\}$

2. Consider the grammar with productions $S \rightarrow a \mid Sa \mid bSS \mid SSb \mid SbS \mid \lambda$.
- Show leftmost derivation, rightmost derivation, and derivation tree for the string "aabbbaaa". 5
 - Find if the given grammar is ambiguous or not. Show the logic behind your claim. 3
 - What are the key properties of parse trees? 2
3. a) Write the formal definition of Grammar. 1
- b) - Construct the regular expression for the given FA. 4



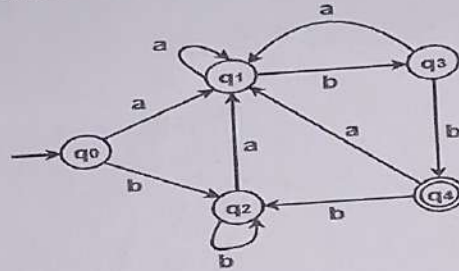
- c) Design an FA from the given regular expression $10 + (0 + 11)0^* 1$. 5

4. a) Convert the following NFA to DFA.



5

- b) Reduce the number of states in the following DFA.



5

5. a) Consider the following rules for a C identifier (c.g. a variable name):
 -an identifier must start with a letter or an underscore
 -an identifier can contain letters, digits, and underscores
 Write a **grammar** that denotes the language containing all valid C identifiers.

2

- b) State two fundamental differences between NFA and DFA.

3

- c) Determine whether the following grammars are simple. Justify your answer.

i) $G_1 = (\{S, E, F\}, \{+, *, \%, a, b\}, S, P)$ with P given as:
 $S \rightarrow +E \mid *F \mid \%F$, $E \rightarrow \%S \mid \%F \mid \lambda$, $F \rightarrow a \mid b \mid \lambda$

ii) $G_2 = (\{S, E, F\}, \{+, *, \%, a, b\}, S, P)$ with P given as:
 $S \rightarrow aA \mid bB$, $A \rightarrow aA \mid baA$, $B \rightarrow bb \mid \lambda$

2

6. a) Write the formal definition of Pushdown Automata.

- b) Consider the NPDA $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{a, b, c\}, \{0, 1\}, \delta, q_0, 0, \{q_3\})$ where δ is given by

2×2.5=5

$\delta(q_0, c, 0) = \{(q_0, 0)\}$	$\delta(q_1, a, 1) = \{(q_1, 11)\}$	$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$	$\delta(q_3, c, 0) = \{(q_4, 0)\}$
$\delta(q_0, a, 0) = \{(q_1, 10)\}$	$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$	$\delta(q_2, \lambda, 0) = \{(q_3, \lambda)\}$	$\delta(q_4, \lambda, 0) = \{(q_3, \lambda)\}$
$\delta(q_0, \lambda, 0) = \{(q_3, \lambda)\}$		$\delta(q_2, c, 0) = \{(q_4, 0)\}$	

Draw the NPDA. Also, show if the string **cccaabb** is accepted or not by M .

- c) Construct a PDA that can be accepted for the language $L = \{a^n b^{2n} : n \geq 1\}$.

3

Metropolitan University, Sylhet
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Spring Semester Final Examination – 2024
Program: B.Sc. in CSE Batch: 57
Course: GED 431: Business Communication

Marks: 40

Time: 2 Hours

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) Describe the “positional power” component of the four components of “influence” and also give an example associated with a daily business communication scenario. 5
(b) If you are a seller of a particular software that processes data on cancer patients and uses AI for early detection techniques, what persuasion strategy will you use for diagnostic centers to buy it? 5
2. (a) In your experience, can you illustrate how an individual can become a charismatic leader by just engaging in conversation? 5
(b) Point out how “storytelling” can be an effective way of business communication for top-level managers. 5
3. (a) “People tend to respond or comply in a scenario where they find similarities. On the basis of your knowledge of “the principle of liking” in the science of persuasion, build an argument. 5
(b) Explain why comparison is a must for anyone to be a charismatic leader. 5
4. (a) Justify how “positional power” can be bypassed by “expertise.” Make sure you give an example. 5
(b) According to the ‘principle of social proof, people follow the lead of similar others; give a discussion of the statement with your agreement or disagreement. 5
5. (a) “The act of integrity makes a person more charismatic than actually having integrity.” 5
What is your opinion regarding the statement?
(b) Reciprocity is the game changer for a new business relations establishment—justify the statement on the basis of “the principle of reciprocity.” 5
6. (a) For modern business communication, story-telling is a must. Now point out what makes a good story. 5
(b) Discuss why “non-verbal signal” is a vital component in the influence game. 5